



CHEMISTRY Department
SIMHADRI

Note No :1



Ref:SMTTP/Chemistry/2025-26/OFC NOTE/28342

Date:

08/04/2025(18:01:04)

Subject:Review OF LMI on OPERATION MAINTENANCE OF SEWAGE TREATMENT PLANT NTPC Simhadri LMI No: LMIOGNENV011

Review OF LMI on OPERATION& MAINTENANCE OF SEWAGE TREATMENT PLANT, NTPC Simhadri LMI No: LMI/OGN/ENV/011

Subject: LMI ON OPERATION& MAINTENANCE OF SEWAGE TREATMENT PLANT.

Above LMI on OPERATION& MAINTENANCE OF SEWAGE TREATMENT PLANTS LMI No: **LMI/OGN/ENV/011** pertaining to Chemistry Section has been reviewed and no changes has been incorporated.

Next Review will be in the Month of March 2027 or if the OGN is revised whichever is earlier.

Put up for the approval of competent authority and for the implementation.

Forwarded for approval please.

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Note No:2

Submitted for review by concern departments for any remark and for approval by CA please.

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Note No:3

It is requested to modify as per standard format of Simhadri LMI (Guidelines for Issue of LMIs as a Controlled Document) LMI/OGN/GEN/002 Rev:04 as uploaded on Simhadri EEMG LAN

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Note No:4

As mentioned in the previous page, changes have been done as per the standard format.

Attached document updated.

May please see and forward further.

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The LMI has been reviewed and found to be in line with the OGN.

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Forwarded please.

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No changes have been proposed in reviewed LMI. Only LMI format changed as per OGN.

Submitted for approval please.

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LOCATION MANAGEMENT INSTRUCTION

OPERATION& MAINTENANCE OF SEWAGE TREATMENT PLANT, NTPC Simhadri

LMI No: LMI/OGN/ENV/011 Rev.2, MARCH-2025

OPERATION& MAINTENANCE OF SEWAGE TREATMENT PLANT, NTPC Simhadri

LMI No: LMI/OGN/ENV/011 Rev.2, MARCH-2025

LMI APPROVAL DOCUMENT

1	LMI NAME	OPERATION& MAINTENANCE OF SEWAGE TREATMENT PLANT
2	LMI No.	LMI/OGN/ENV/011
3	Reference Document	COS-ISO-00-OGN/ENV/011
4	Superseded Document	LMI/OGN/ENV/011, Revision:2 Revision Date: 27.03.2025
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8	Approved By	Shri Sameer Sharma ED (Simhadri)
9	Date of Issue	27.03.2025
10	Index List	Enclosed
11	Revision Index	Enclosed



OPERATION& MAINTENANCE OF SEWAGE TREATMENT PLANTS INDEX

Sl. No.	Contents	Page No
1.0	Introduction	1
2.0	Superseded Document	1
3.0	Scope	1
4.0	Objective	1
5.0	Operational problems observed at STP	1-2
6.0	Process	2-5
7.0	Secondary treatment by activated sludge process	5-8
8.0	Effluent standards	8-9
9.0	Specific instructions to be followed	9-10
10.0	Startup of sewage treatment plant	10
11.0	Regular Operation of STP	11-12
12.0	Do's & Don'ts	12
13.0	Process monitoring& Trouble Shooting	12-18
14.0	Safety	18
15.0	Reference	18
16.0	Review	18
	Annexure-1 Parameters to be maintained	19
	Annexure-2 Abbreviations	19
	Annexure-3 Test procedures	20-21

Guidelines for Operation of Sewage Treatment Plants (With Extended Aeration Technology)

1.0 INTRODUCTION:

Sewage treatment plants are essential to maintain environmental quality and to protect public health. The Sewage Treatment Plant (STP) includes physical, chemical and biological processes to remove physical, chemical and biological contaminants.

NTPC Simhadri has two Sewage treatment plants. One is in Township for the treatment of domestic effluents with a capacity 1700 m³/day and second plant is located in Main plant premises for the treatment of Toilet effluents in the plant with a capacity 96 m³/day as per environmental guidelines before discharge into the water body. At NTPC Simhadri, sewage treatment plants are installed with extended treatment technology involving Activated Sludge Process (ASP).

2.0 SUPERSEDED DOCUMENT - LMI/OGN/ENV/011, Issue No: 2, Revision:2

3.0 SCOPE

Applicable to Sewage treatment plants with extended treatment technology located at NTPC Simhadri.

4.0 OBJECTIVE:

Objective of sewage treatment or domestic wastewater treatment is to produce treated effluent by the process of removing contaminants from sewage and domestic wastewater, as per prescribed environmental norms by the State Pollution control boards in the water consent conditions. The principal objective of wastewater treatment is generally to allow human and industrial effluents to be disposed of without danger to human health or unacceptable damage to the natural environment.

Sewage treatment plants are operated under optimized operational conditions thus paving the way for always getting the constant effluent quality. The consistent better effluent quality will also encourage its reuse for other purposes in the plant premises leading to zero discharge of effluent.

5.0 OPERATIONAL PROBLEMS OBSERVED AT STP:

The various prominent problems causing improper treated effluent quality are:

1. Bad odour

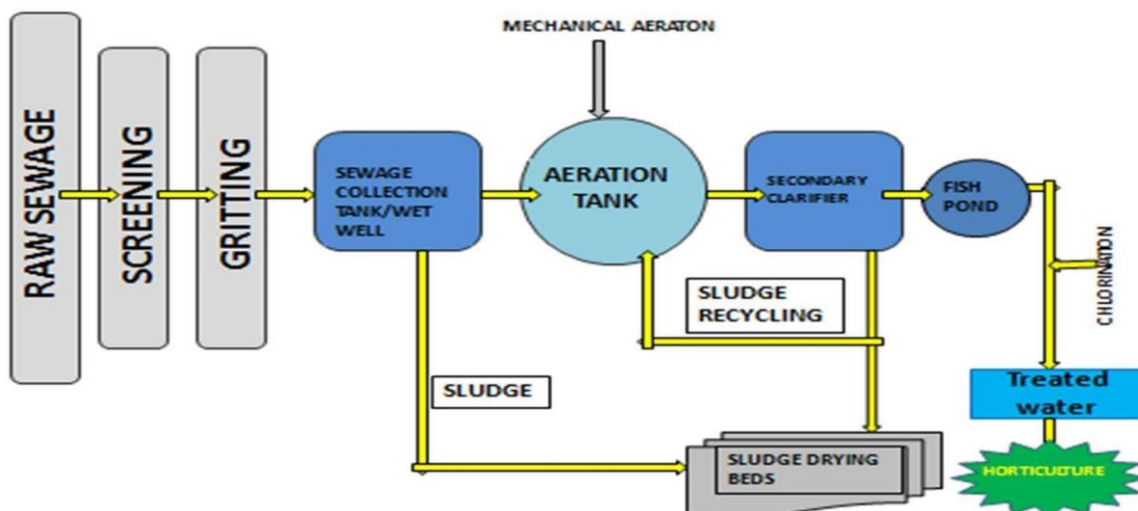
2. Foaming
3. High Biological Oxygen demand (BOD)
4. High Chemical Oxygen demand (COD)
5. High Total Suspended Solids (TSS)

Lack of operational awareness also affects the aesthetics and treated effluent quality making it difficult for its reuse.

6.0 PROCESS:

A) TOWNSHIP STP: The Raw sewage will be passing through Screen & grit chamber to trap the free-floating material & further will be drained into the Sewage collection tank (Wet well). The collected Sewage is pumped into an aeration tank equipped with surface aerators (Mechanical aeration) providing necessary oxygen for biodegradation of organic matter. The biodegraded sewage from aeration tank is then taken to the secondary clarifier for settling of bio sludge. The overflow from secondary clarifier is taken into the fishpond & then in the outlet, optimum bleaching powder is dosed for disinfecting purpose. The treated water is used for horticulture in the township. The settled sludge from secondary clarifier is then drained to sludge drying beds, where a part of sludge is recirculated to aeration tank.

The typical scheme of the process is as follows.

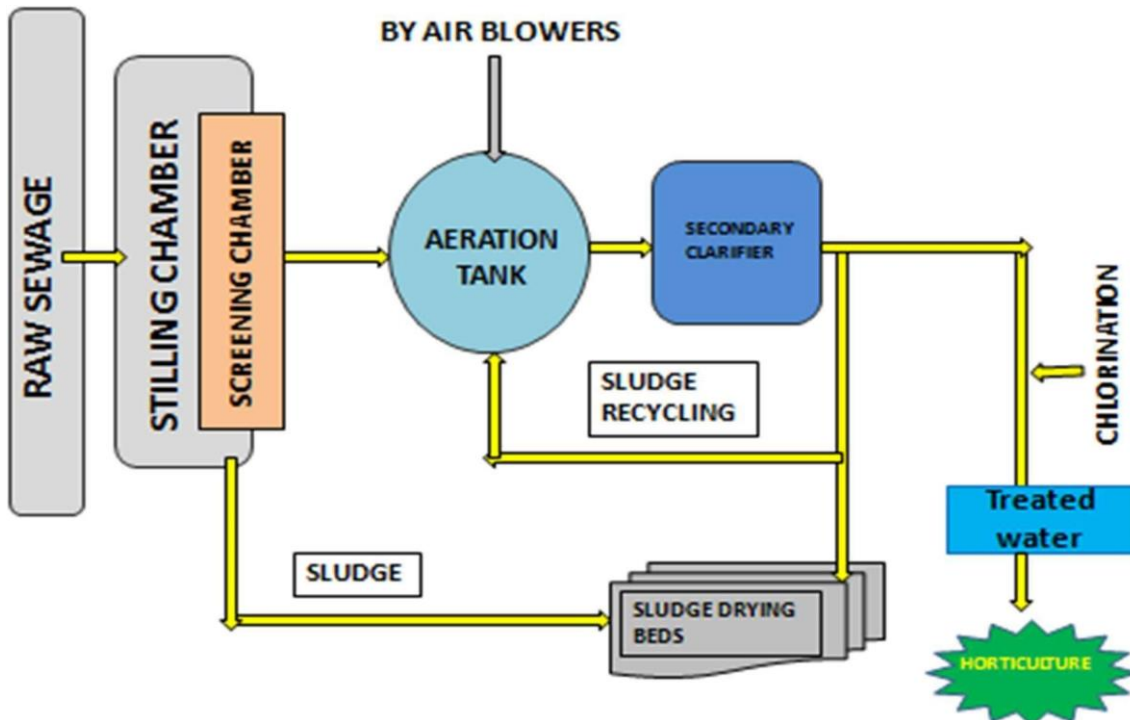


B) MAIN PLANT STP:

The Raw sewage will be passing through the screen chamber with suitable crib at the inlet of stilling chamber. The outlet from the stilling chamber is fed into the aeration tank of the extended aeration type. Aeration tank is provided with fine/medium diffuser type arrangement to distribute air uniformly throughout the aeration tank for biodegradation of organic matter. The bioflocs formed in the aeration tank with other suspended particles

will be settled in the clarifier. From here the settled sludge will be recycled back to the aeration tank where required MLSS is maintained to facilitate the oxidation of dissolved organics. The excess sludge generated will be pumped to the sludge drying bed for drying before disposal. Chlorination is done with bleaching powder in the outlet water to disinfect the water as well as to oxidize the trace organics. The final treated water is disposed or used for horticulture.

The typical scheme of the process is as follows.



6.1 PRIMARY TREATMENT: SEPARATION OF SOLIDS

First, the solids are separated from the wastewater stream. This process is basically primary sedimentation. The raw wastewater enters the treatment plant and is immediately passed through a bar screen. This removes the largest solids from the water. The sewage then passes through Grit removal system followed by sedimentation tanks. Most of the solids settle to the bottom of the tanks and form a watery sludge, known as 'primary sludge', which is removed for separate treatment. The sewage remaining after settlement has taken place, is known as 'settled sewage'.

6.2 SECONDARY (BIOLOGICAL) TREATMENT

6.2.1 SEPARATION OF DISSOLVED ORGANIC MATTER

Settled sewage then flows to a biological treatment stage in aeration chamber where it meets micro-organisms which remove most of the remaining organic pollutant by activated sludge process. Dissolved biological matter is progressively converted into a solid mass by using water-borne micro-organisms.

6.2.2 FINAL SETTLEMENT OF SLUDGE

After the wastewater leaves the aeration chamber, it enters the clarifier. The clarifier is a type of sedimentation basin in which the heavier solids sink to the bottom and the lighter materials float to the surface. One of the primary purposes of the clarifier is to remove the microorganisms from the water. After digesting organic matter in the aeration basin, these microorganisms now have food, grit, and other particles stuck to their outer enzyme coating. So, they are heavy and sink to the bottom (or floc out) in the clarifier. The supernatant, a clear liquid, rises to the top and is allowed to flow out of the clarifier. In the case of the activated sludge process, some of the secondary sludge is returned to the aeration tanks for further contact with the sewage. The secondary sludge from biological treatment also requires separate treatment and disposal and may be combined with the primary sludge for this purpose.

6.2.3 SLUDGE TREATMENT

Sewage treatment also produces sludge as by-product, and these must be treated and disposed of separately in an economical and environmentally acceptable way. Finally, the biological solids are neutralized then disposed of or re-used, and the treated water may be disinfected. All methods of sewage treatment generate organic sludge (or 'bio-solids') as by-products, and these must be managed separately from the liquid sewage. Raw (untreated) sludge has a very high oxygen demand and must not be allowed to enter the water environment. There is, therefore, a need to deal with them in a way that permits their ultimate disposal in an environmentally acceptable and sustainable manner. The sludge disposal 'route' selected for a given sewage treatment works will depend on several factors including its location, the availability of suitable farmland, the characteristics of the sludge and the overall cost.

6.3 TERTIARY TREATMENT

6.3.1 DISINFECTION OF TREATED WATER

After secondary treatment, the supernatant is chlorinated and allowed to sit in a contact chamber while the chlorine reacts with microorganisms in the water. This process disinfects the water, killing the disease-causing microorganisms. The final effluent can be discharged into a stream, or it can be used for the irrigation or horticulture purposes. In simhadri the final effluent is used for Horticulture.

6.3.2 APPLIED TO FINAL USE

If it is sufficiently clean, it can also be reused back into the system as raw water for industrial purposes in future. In circumstances where the highest quality of effluent is

required, a third (tertiary) stage of treatment can be used to remove most the remaining suspended organic matter and other contaminants, like heavy metal & phosphates from the effluent before it is reused. Tertiary treatment is affected by sand filters, mechanical filtration, activated carbon filters, UF & RO depending on the requirement of end use.

7.0 SECONDARY TREATMENT BY ACTIVATED SLUDGE PROCESS:

7.1 BIOLOGICAL GROWTH:

The extended aeration process provides biological treatment for the removal of biodegradable organic wastes under aerobic conditions. Air is supplied to provide the oxygen required to sustain the aerobic biological process. Mixing is provided by aeration or mechanical means to maintain the microbial organisms in contact with the dissolved organics.

The bacterial growth in a batch culture generally has five distinct phases, namely,

7.1.1 Lag phase:

The lag phase represents the period required for bacteria to acclimatize to their new environment.

7.1.2 Log growth phase:

During log phase, excess substrate is available to bacteria and rate of their growth is a function of the ability of the bacteria to metabolize the food.

7.1.3 Declining growth phase:

During declining growth phase, bacterial growth rate decreases gradually as the food supply gets exhausted.

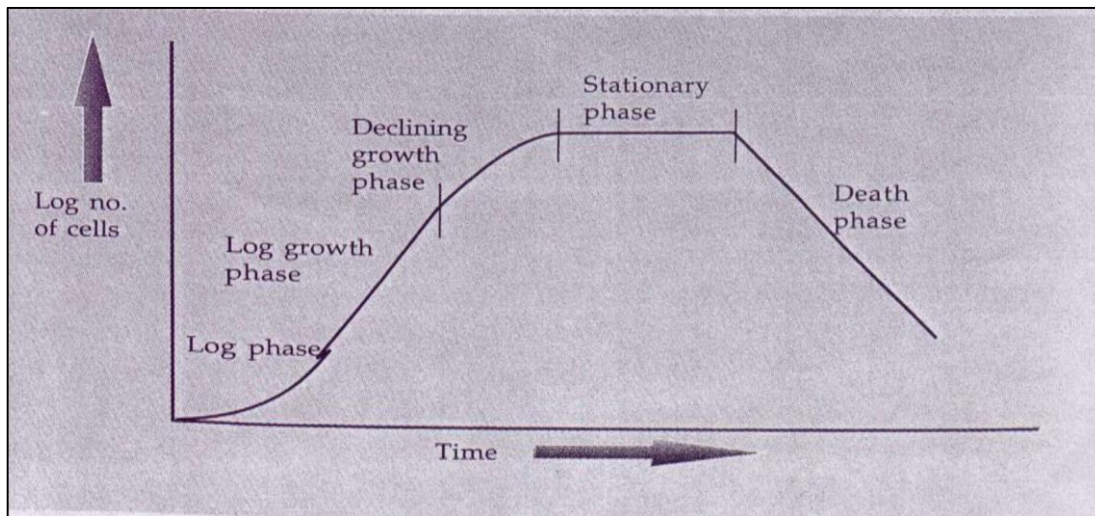
7.1.4 Stationary phase:

In the stationary phase the growth rate equals the death rate.

7.1.5 Death phase.

During death phase availability of food is at minimum and the bacteria are forced to metabolize their own protoplasm hence bacteria death rate exceeds the production of new bacterial cells.

These phases are schematically shown in Figure 2 below:



Days Figure 2: Growth pattern of bacteria in a batch culture

In a continuous culture system, such as a biological wastewater treatment reactor, the microorganisms are made to function at a position of choice on the growth curve by regulating the microorganism mass in the reactor in relation to the incoming food supply. The position of choice influences the process efficiency for removal of organic waste, excess sludge production, and oxygen and nutrients requirements. It also determines if the excess sludge would require further treatment to improve its drain ability.

7.2 EXCESS SLUDGE WASTING:

Excess bio-sludge is commonly wasted from return sludge line. It can also be wasted directly from aeration tank. If excess bio-sludge is directly wasted from aeration tank, then increased volume of sludge is a disadvantage. However, if excess bio-sludge is mixed with influent of primary settling tank and wasted as mixed sludge of primary settling tank, then direct wasting from aeration tank has no influence on final volume of sludge and therefore can easily be adopted.

7.3 RETURN SLUDGE FLOW:

Sufficient return sludge capacity should be provided if the biological solids are not to be lost in the effluent. However, a return flow rate higher than that is required unnecessarily increases solids loading on settling tank and results in withdrawal of dilute sludge. The ratio of return sludge flow to average flow can be set based on Sludge Volume Index (SVI). SVI is defined as the volume in ml occupied by one gram of activated sludge Mixed Liquor Suspended Solids (MLSS), dry weight, after settling of 30 min in a 1000 ml graduated cylinder. This can therefore be shown with the following equation:

$$SVI = \frac{\text{Total Volume occupied by sludge in ml}}{\text{Dry weight of sludge in that volume}}$$

In practice, the ratio of return flow (QR) to average influent (Q) can be decided by following relationship:

$$QR/Q = MLSS / (106/SVI - MLSS)$$

It is seen that for usually encountered SVI values between 50 and 200, the MLSS in the aeration tank can be maintained between 4000 to 6000 mg/L by adopting QR / Q value of 0.5 and 1.0. A higher return for poorly settling sludge is for maintaining the same MLSS concentration. In the absence of a return flow rate measurement facility, low flows are generally attained by restricting the flow rate with the help of valves thereby causing unnecessary waste of energy.

Existence of high return flow installed capacity accompanied by absence of return flow measurement facility can, at times, lead to excessive return flow rate even without knowledge of operator, which, in turn, can totally upset the operational control of the process. Thus, flow-measuring facility for the return flow is very essential for process control. Since the sludge wasting is also done mostly from the return flow line, this facility may also fulfill the requirement of measurement of excess sludge flow.

7.4 ROLE OF SOLID SEPARATION FACILITY

Settling tank performs both clarification for treated water and thickening functions for sludge. Settling tank is an integral part of the activated sludge process and design of a reactor cannot be considered independently of the design of associated settling facility. It is essential that the mixed liquor solids be separated very effectively in the settling facility so to meet the discharge standards. Consideration of settling properties of sludge as related to thickening requirement for proper plant operation, surface overflow rates, solids loading and flow through velocity is very important in the design of solid separation facility.

Secondary settling facility must perform two functions, namely, clarification of liquid and thickening of sludge. The first function involves separation of the mixed liquor suspended solids from treated wastewater to provide clarified effluent and the second function involves transportation of solids to the bottom of clarifier to provide thickened return sludge. Area requirement for clarification is based on the settling velocity of smallest particle to be removed and is derived from column settling tests for sludge concentration at which the plant is to operate.

7.5 OXYGEN REQUIREMENT

The theoretical oxygen requirement is related to the organic matter oxidized. The portion of organic matter synthesized into microorganisms, which is wasted from system, is not oxidized and thus does not require any oxygen. The oxygen equivalent of microorganisms wasted from the system is estimated as 1.42 kg oxygen per 1 kg of bacterial cell.

Therefore, net oxygen requirement in a process can be computed as:

$$O_2 \text{ (Kg/day)} = 1.47[\text{mass of BOD}_3 \text{ removed (Kg/day)}] - 1.42 [\text{mass of organisms wasted (Kg/day)}]$$

Where, 1.47 is a factor to convert BOD₃ to ultimate BOD.

It is required that a minimum Dissolved Oxygen (DO) concentration of 1 to 2mg/L should be maintained throughout the aeration tank to keep the reactions aerobic.

7.6 HYDRAULIC RESIDENCE TIME (HRT): Hydraulic residence time refers to the wastewater liquid within the treatment plant. Hydraulic residence time is the amount of time it takes a particle of water, upon entering the plant to leave the plant. In activated sludge plants, HRT is measured in hours, usually between 20-30 hours, under normal flow conditions.

7.7 SLUDGE AGE: Sludge age refers to the wastewater solids within the treatment plant. Sludge age is the amount of time a particle of (activated) sludge remains in the treatment plant, settling in the clarifier and returning to the aeration basin. This cycle happens many times before that sludge particle is wasted out of the system. Sludge age is thus measured in days, usually 3 to 30 days, depending on the type of activated sludge plant and wasting rate.

7.8 CELL RESIDENCE TIME: The solids retention, or mean cell residence time, is the key operating parameter because it directly controls nearly all other parameters (equations) of interest when operating a biological treatment system. The Average Length of Time in Days that an Organism Remains in the Secondary Treatment System. Cell Residence Time (CRT) in Days= (Total MLVSS, m3)/ (Total MLVSS Wasted, m3/d)

S.No	Parameter	Conventional Activated Sludge	Extended Aeration Activated Sludge
1	Aerator Detention Time	4-8 Hrs	16-24 Hrs
2	F:M Ratio	0.25-0.45	0.05-0.15
3	CRT	4-6 days	15-25 days

8.0 EFFLUENT STANDARDS:

The statutory limits fixed by pollution control boards which are given as below.

Parameters	Units	Limits for final treated effluent as per NGT Guidelines dated 30.04.19
pH	--	5.5 - 9.0
Total Suspended Solids (at 103 – 105 °C)	mg/l	20
Biochemical Oxygen Demand (BOD 3 at 27 °C)	mg/l	10
Chemical Oxygen Demand (BOD at 27 °C)	mg/l	50
Nitrogen- Total	mg/l	10
Faecal Coliform	MPN/100ml	Desirable-100 Permissible-250

Phosphorous–Total (For discharge into ponds, lakes)	mg/l	1
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9.0 SPECIFIC INSTRUCTIONS TO BE FOLLOWED

S.No	Activity	Responsibility
9.1	INSTRUCTIONS FOR START-UP OF THE PLANT:	
9.1.1	PREPARE BIOLOGICAL SEED IN THE AERATION TANK	HOD(Civil)
	While commissioning the plant or restart of the plant after long shutdown, acclimatized micro-organisms in cow dung (biological seeds) are to be used and should be maintained in aeration tank by the following steps:	
	a) Clean the aeration tank,	
	b) Remove the extraneous materials like straw, grass, etc. from naturally available bio matter, cow dung.	
	c) Slowly add cow dung into the aeration tank in uniform proportions and small doses at 100-150 lit/hr (approx. 10 buckets/hr).	
	d) Cow dung added should be approximately equal to 1/10th of the aeration tank volume,	
	e) Continuously operate the surface aerator so that the mixture of cow dung and water is thoroughly mixed and oxygen from the air is provided continuously.	
	f) Do not allow any extra water to flow inside the aeration tank.	
	g) The process of seeding will take about 5 to 30 days' time depending upon the temperature and humidity of the atmosphere.	
	h) Run the sewage treatment plant	
	i) After the seeding period a brownish colored sludge is formed which is easily settle able.	
9.1.2	INSTRUCTIONS FOR CHARGING OF AERATION TANK	HOD(Civil)
	To start the plant ensures that Screen chamber and before that screen chamber is clear from the debris and plastic materials. Similarly, all the tanks should be cleaned and free from any waste material before sewage is taken in to the plant.	
	The sewage passes through the grid chamber and v notch chamber before entering the aeration tank. The flow rate of the sewage entering the aeration tank point should be monitored by measuring the height at the V-notch.	

S.No	Activity	Responsibility
9.2	INSTRUCTIONS FOR MAINTAINING OPERATIONAL PARAMETERS	
9.2.1	The following conditions are to be maintained to ensure that the micro-organisms present in the floc or activated sludge is in a state of proper growth and, therefore, working at maximum efficiency.	HOD (Civil Maintenance)
A	A continuous supply of sewage with a uniform and nutritionally adequate organic content. A highly variable organic load to the plant would be undesirable.	
B	A complete mixing of incoming wastewater and the micro-organisms present in the aeration basin (diffused aeration).	
C	A continuous supply of dissolved oxygen as supplied by the aerator/blower (diffused aeration).	
E	A settling tank (clarifier) where separation of mixed liquor solids from the liquid carrier is completely and efficiently carried forth.	
F	Excess sludge generated by the process should be removed based on MLSS quantity or F/M (Food to Microorganisms) ratio	
9.2.2	Measurement of MLSS once a month is to be ensured	HOD (Chemistry)
9.2.3	MLSS concentration in aeration tank should be adjusted as per SVI	HOD (Civil Maint)
9.2.4	Sludge Recirculation in Extended Aeration Process is to be ensured to maintain the proper MLSS value and necessary biomass in the oxidation pond to speed up the reaction	HOD (Civil Maint)
9.2.5	A minimum Dissolved Oxygen (DO) concentration of 1 to 2 mg/L should be maintained throughout the aeration tank to keep the reactions aerobic. Mechanical Aerator should be in service during operation of STP	HOD (Civil Maint)
10.0	OPERATING INSTRUCTIONS FOR STP WITH EXTENDED AERATION TECHNOLOGY	
10.1	Screening: The screens should be inspected and cleaned daily.	HOD (Civil Maint)
10.2	Removal of Grit: Grit channels have been provided for sedimentation of grit from the sewage. However, since the incoming sewer will first deliver the sewage into Screen Chamber No.1, some grit may get deposited at	HOD (Civil Maint)

	the bottom of this chamber. Similar deposition may also take place at the bottom of the wet well of sewage pumps and screen Chamber No.2. Grit deposited at these places should be removed manually once a day during the period of lean flow.	
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S.No	Activity	Responsibility
	a) Only one of the two channels will operate at a time. The idle channel will be cleaned under no-flow condition. Grit will be removed through sluice valves provided at the bottom of the grit channel and disposed of by burial.	HOD (Civil Maint)
	b) Grit channel should be operated in 24 hours cycles. Thus, grit will be removed at intervals of 24 hours.	-do-
10.3	Aeration of sewage in aeration tanks: The aerator is to be operated continuously, 24 hours every day.	-do-
10.4	Adding activated sludge: Activated sludge withdrawn from the settling tank will be used for the purpose of maintaining process parameters by recirculation in aeration tank.	-do-
10.5	The activated sludge should be added in such quantities that F/M value is in the range of 0.05 to 0.1. MLSS concentration in aeration tank should be adjusted as per SVI. This will be achieved if sludge addition is maintained with the help of running of sludge pumps.	-do-
10.6	Sludge pumps: Sludge pumps will have to be operated continuously. This is because MLSS in the aeration tank will be controlled by operation of sludge pump. After the desired MLSS has been reached, sludge will have to be diverted to sludge drying beds. The operator stationed in the sludge pump house will obtain directions from the chemist and arrange diversion of sludge to drying beds accordingly. The sludge pump should be worked for this purpose. Head over the V-notch should be measured at intervals, by which it will be possible to monitor the inflow of sludge into the aeration tank.	-do-
10.7	Sludge drying beds (SDB)	
	a) Excess sludge from the clarifier can be discharged to the SDB at required intervals so that the concentration of MLSS is maintained in the aeration tank. Wasting of excess sludge may have to be done from the system daily.	-do-

b)	The sludge application is maintained at 0.15 m given at least seven days drying period. The dried sludge is removed and is used as manure.	-do-
10.8	Disposal of final effluent: The final effluent from the settling tank will overflow into the launder and will be led to treated sewage sump.	-do-

S.No	Activity	Responsibility
10.9	Chlorination:	
a)	Treated sewage from outlet launder of clarifier is collected in the treated sewage sump which is designed to provide 30-minute contact time for chlorination.	HOD (Civil Maint)
b)	Chlorination of treated sewage effluent is to be done by bleaching powder dosing. The residual chlorine in the final effluent is to be maintained approx. 0.2-0.5 ppm.	-do-
10.10	Tests to be performed: Samples of incoming sewage, mixed liquor in aeration tank, final effluent should be taken, and the tests are to be performed as per annexure 1	HOD (Chemistry)
10.11	It must be ensured that unless every piece of equipment, all sections of the plant and the environment are kept clean at all hours, successful function of the plant cannot be accomplished.	HOD (Civil Maint)
	All the required operational parameters are to be met to ensure effluent quality	-do-
11	Operation of the mechanical equipment:	
11.1	This treatment plant should be run round the clock without stoppage. Septic condition may develop if the plant is given rest. Once this condition develops, it will take quite some time to bring it back to normal. When the Septic condition develops the sewage should be aerated heavily until brown color returns in the aeration tank.	HOD (Civil Maint)
11.2	For making the plant always run, the necessary preventive maintenance schedule should be followed. The equipment which are critical for operation and require periodical attention are: a) Sewage pumps b) Aerator c) Sludge pumps d) Circulating drive of secondary settling tank and associated accessories and e) Treated water pumps.	HOD (Offsite-Mechanical Maint)

11.3	Preventive maintenance schedule & operating instructions for this equipment should be made. It is desirable that logbooks be maintained for operating pumps and equipment	HOD (Civil Maint)
11.4	The STP effluent after achieving the desired effluent quality can be used for horticulture purpose. For zero discharge, the balance water if any can be utilized as process water after further treatment of treated sewage water to desired BOD levels and turbidity levels.	-do-

S.No	Activity	Responsibility
12	DO'S & DONT'S to be ensured	
12.1	DO'S	
a)	Clean all the units for debris, waste, material, nuts, wires or any extraneous material.	HOD (Civil Maint)
b)	Screen chamber and grit chamber to be cleaned manually every day during the lean flow period.	-do-
c)	To check healthiness, stand by pump should be operated, once per day.	-do-
d)	Preventive maintenance schedule & operating instructions for the equipments required for STP should be made and followed.	-do-
e)	Sludge should be recycled to aeration tank and sludge wastage should be done based on results of obtained from MLSS test. Sludge can be sent to drying beds when MLSS >4000 mg/l.	-do-
f)	Sludge applied on the beds for drying should not be more than 0.15 m in height. Sludge removal from bed should be carried out in every seven days gap.	-do-
g)	Filtrate of SDB should be pumped back into the screen chamber cum wet well and dry well.	-do-
h)	The surrounding area of STP may be given an aesthetic look by way of planting flowering plants and developing greenery.	-do-
12.2	DONT'S	
a)	Do not allow floating plastic materials etc. into the raw effluent sump otherwise it will choke the pump strainer.	-do-
b)	Do not allow surface aerator to run when liquid level has gone below the tip of the cone of the aerator.	-do-
c)	Do not allow the floating plastic material etc., into the aeration tank otherwise it will damage the impeller of the sewage aerator.	-do-
d)	Do not start the pump without ensuring water in the priming compartment. Do not allow the pump to run dry. Do not run standby pumps simultaneously.	-do-

e)	Do not allow the washing out of the suspended solids from clarifier. This can be checked by SVI and sludge level in the clarifier.	-do-
f)	Sludge level in clarifier should not raise 1m above the bottom level.	-do-

13.0 PROCESS MONITORING AND TROUBLE SHOOTING

13.1 There are three major control mechanism for the activated sludge treatment process

1. Aeration rate,
2. Return sludge rate, and
3. Waste sludge rate.

Control of these three variables in addition to providing the proper environment (physical, chemical, biological and nutritional requirements) all lead to good sludge quality. There are many different techniques to activated sludge process control, all the methods are directed towards three common goals.

- First to provide the proper environment in the system.
- Secondly, to balance the mass of solids under aeration with the organic loading
- Finally, there must maintain a balance return sludge rate which allows mixed liquor thickening in the secondary clarifiers, yet also removing the MLSS rapidly enough to prevent septic sludge conditions.

13.2 PROCESS MONITORING

Monitoring the process is essential to successful process control. Organized and proper sampling, measurement, and data collection is used to make proper control decisions. Monitoring tests are used to evaluate the performance of the system. Monitoring tests include effluent quality test such as BOD and TSS. Process control test include observations of the aeration tank and secondary clarifiers, flow data, solids concentrations, sludge quality data, and environmental data. In addition, calculated data is required for aeration, return sludge and waste sludge. Data for plant process control and monitoring might include the following.

A. Flow:

- Influent/Effluent
- Return RAS
- Waste (WAS)
- Air
- Recycle Flows

B. BOD and COD: - Influent/Effluent

C. TSS: -influent/Effluent

OBSERVATIONS:

- a. Aeration Tank - Spray patterns - Flow patterns - Turbulences - Color - Odor - Foam
- b. Secondary Clarifiers - Clarity - Bulking - Washout - Clumping/Ashing - Straggler floc - Pin floc
- c. Solids concentration: - Mixed liquor - Return sludge - Waste sludge - Reaeration (for contact stabilization)
- d. Sludge settling test data: - Settled Sludge Volume (SSV) - Settled Sludge Concentration (SSC) - SVI - Settling velocity - Settling curves.
- e. Sludge blanket depth (for each clarifier)
- f. Dissolved Oxygen DO: - Aeration tank (inlet, midpoint, outlet - Reaeration tank (for contact stabilization)
- g. Microscopic enumeration and characterization: - Mixed liquor - Return sludge
- h. Turbidity - Secondary effluent
- i. Heavy metals - Influent/effluent
- j. SOUR (specific oxygen uptake rate) - Aeration tank inlet/outlet (mixed liquor) - Fed and unfed samples.
- k. Aeration Tank and Clarifier Oxygen Reduction Potential (ORP)
- l. Nitrogen (ammonia TKN, nitrite, nitrate) - Influent/effluent
- m. Phosphorus - Influent/effluent
- n. Alkalinity - Influent/Effluent

Note: An alkalinity drop of more than 30 mg/L as CaCO₃ usually indicates nitrification.

As can be seen, there are many processes evaluation and control test available to monitor and control the activated sludge system. The operator must decide which test are essential, how often to test, where to test and the sample type. Once the test has been selected, the operator must then identify common ranges or targets for the test data collected.

13.3 Troubleshooting is a step-by-step procedure to diagnose and correct a problem.

Troubleshooting consists of seven major activities:

- a. Recognizing that a problem exists.
- b. Assigning priorities to ensure that each problem receives adequate attention.
- c. Gathering facts in an organized manner to establish base data.
- d. Finding the cause of the problem as quickly as possible to minimize the impact on process and cost.
- e. Identifying possible alternative courses of action.
- f. Recording the facts in an organized manner.
- g. Eliminating or reducing the cause of the problem to minimize recurrence.

13.3.1 The first step in troubleshooting is to recognize that there is a problem. All plants should have Standard Operating Procedures (SOP's) which include procedures for

operators to detect problems. The ability to recognize minor problems and to correct them early can prevent them from developing into major problems.

13.3.2 Once a problem is detected, an assessment must be made as to its priority over other existing problems. The three major factors in assigning priorities are seriousness, urgency, and growth of the problem in terms of impact on process performance or safety of personnel or equipment.

13.3.3 Gathering facts is a four-step process: stating the problem, defining the boundaries, noting differences and changes, deciding on cause/effect.

Problem statement. The problem statement should be a precise description of the problem in terms of observable facts.

Boundary definition. Ask the following types of questions: Where is the problem observed? When was the problem first detected? What is the extent of the problem?

Differences and Changes. Note any difference or changes from what is normal.

Cause/effect. Identify probable cause.

13.3.4 Possible alternative courses of action may be from the following five categories: contingent, interim, corrective, adaptive and preventive.

Contingent actions. These are taken immediately to minimize the effect of the problem.

Interim actions.

Interim actions are taken while the problem is being solved and repairs are being made.

Corrective actions. Corrective actions address the cause of the problem.

Adaptive actions. Adaptive actions allow the plant to live with the problem.

Preventive actions. These reduce or eliminate the cause of a problem.

13.3.5 Procedure needs to be established to ensure that facts are recorded so that:

Important facts are not lost when a problem occurs.

The cause of a problem is more quickly isolated.

Management is provided with the records.

Engineering is provided with the data to analyze alternatives.

13.3.6 The establishment of a good process control strategy and monitoring program is essential to troubleshooting the activated sludge process. Before troubleshooting a problem, it is important to review the plant's physical capabilities and limitations. A design audit should be conducted. The design audit should include:

1. Plant loading (a. Organic b. Hydraulic c. Industrial d. Toxic e. Seasonal variations f. Infiltration/inflow g. Return process streams)
2. Plant layout
3. Flow measurements
4. Bar Screens
5. Comminutors
6. Grit Removal
7. Primary Clairfiers
8. Aeration a. Hydraulic detention time b. BOD loading c. Oxygen availability.
9. Secondary clarifier (a. Surface overflow rates b. Solids loading rates c. Depth at weirs d. Possible RAS flow range e. Short circuiting)
10. Solids handling

13.3.7 Operational problems which are commonly experienced are:

- i. Aeration. The operator should observe the entire aeration tank for turbulence. The extent of turbulence will indicate whether all sewage, return sludge, and mixed liquor are thoroughly mixed throughout the tank.
- ii. Foaming. The type of foam or scum, if any, accumulated over the aeration tank surface and to a lesser extent the color of the mixed liquor reveals process status and indicates generalized long-term sludge wasting requirements.
- iii. Solids Washout. Excessive solids washout over the final effluent weirs, when the upper surface of the sludge blanket is more than three feet below the clarifier surface and when the sludge settles properly in the laboratory, should not be confused with classic sludge bulking. Solids washout, which differs from classic sludge bulking, is more frequently caused by hydraulic overloading, toxic substances (such as high pH, heavy metals, phenols, etc.) or inappropriate secondary clarifier design.
- iv. Bulking. Such conditions are evidence by a homogeneous appearing sludge blanket that extends throughout the entire clarifier, and can be observed at the surface while the mixed liquor solids pour over the final effluent weirs.
- v. Clumping/rising sludge. At times, large masses of sludge, possibly one foot in diameter, may be seen rising, then bursting, and finally spreading over the clarifier surface.
- vi. Cloudy secondary effluent. Secondary effluent is cloudy and contains suspended matter.
- vii. Ashing. Small particles of ash-like material floating on the surface of the clarifier and in mixed liquor settleability test.
- viii. Pinpoint floc. Very small, compact pin floc, usually less than on thirty-second of an inch in diameter, may be observed suspended throughout moderately turbid final clarifier tank contents.

ix. Straggler floc. Small, almost transparent, very light fluffy, buoyant sludge particles (one eighth to one-quarter inch in diameter) may be observed rising to the clarifier surface near the outlet weirs.

13.4 CORRECTIVE ACTIONS: Some causes and corrective actions for filamentous growth, which generally observed operational problem in activated sludge process are below.

Cause - low dissolved oxygen

Correction - increase dissolved oxygen and maintain at least 1.5 to 2 mg/L

Cause - nutrient deficiency:

Correction - add nutrients to achieve a ratio of 100 carbon:5 nitrogen:1 phosphorus: 0.5

Iron

Cause - high sludge age (excessive MLSS) and low F/M ratio):

Correction - increase sludge wasting to reduce the sludge age and increase the F/M ratio

Cause - very young sludge (extremely low MLSS): Not enough organisms to handle the incoming load. This causes stabilization to occur in the final clarifier with dissolved oxygen going to zero. This will favor filaments growth.

Correction - build MLSS by not wasting, or haul in "seed" sludge from another plant

14. SAFETY:

14.1 Personal safety: List of various safety considerations that are important when working in an activated sludge plant.

1. Falling into tanks, especially aeration tanks where currents can pull you under the water c. surface.
2. Noise
3. Exposure to waterborne and bloodborne pathogens
4. Rotating equipment
5. Electrical hazards
6. Slippery surfaces
7. Confined spaces
8. Compressed air
9. Chemicals and chemical equipment

Operators should follow all federal and state safety requirements. Safety programs and emergency procedures should be in place and always followed.

14.2 General safety: Discuss procedures for entering treatment tanks or vessels.

Owners of wastewater treatment facilities should clearly define all confined spaces.

Operators should know them and follow all confined space entry procedures.

15: REFERENCE: Guidelines for Operation of Sewage Treatment Plants (With Extended Aeration Technology) COS-ISO-00-OGN/ENV/011 Rev. No.: 2 Dec. 2022

16.0 REVIEW: GM (O&M) is responsible for reviewing this LMI document, on a two-yearly basis or as necessary.

Annexure-1

Sample	Parameters	Unit	Acceptable operating range	Frequency	Record Availability
Influent (STP Main plant)	pH	-	-	Monthly basis	water Laboratory/Intranet
	TSS	mg/L	-	Monthly basis	
	BOD	mg/L	-	Monthly basis	
	COD	mg/L	-	Monthly basis	
Treated effluent (STP main plant & Township)	pH	--	6.5-8.5	Weekly basis	water Laboratory/Intranet
	Cond.	µs/cm	-	Weekly basis	
	TSS	mg/L	<20	Weekly basis	
	BOD	mg/L	<10	Weekly basis	
	COD	mg/L	<50	Weekly basis	
Mixed Liquor	MLSS concentration	mg/L	4000-6000	Weekly basis	water Laboratory/Intranet
	SVI		50-100	Weekly basis	

Annexure-2**ABBREVIATIONS USED**

ASP	Activated Sludge Process
BOD	Biological Oxygen Demand
COD	Chemical Oxygen demand
Cond.	Conductivity
CPCB	Central Pollution Control Board
DO	Dissolved oxygen
F/M	Food to Microorganisms ratio
MLSS	Mixed Liquor Suspended solids
MLVSS	Mixed Liquor Volatile Suspended Solids
NETRA	NTPC Energy Technology Research Alliance
ppm	Parts per million
SDB	Sludge Drying Bed
SPCB	State Pollution Control Board
STP	Sewage treatment plant
SVI	Sludge Volume Index
TDS	Total dissolved solids

TSS	Total suspended Solids
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ANNEXURE- 3

Test procedures

1. Mixed Liquor Suspended Solids (MLSS)

Sample size: 100 ml.

Apparatus used in the analysis:

- i) Gooch Crucible/Membrane
- ii) Filtration apparatus
- iii) Oven

A) PROCEDURE:

- i) A well-mixed 500-1000 ml sample is taken from the aeration tank of STP.
- ii) Weigh 47mm dia (0.45 µm) cellulose nitrate membrane or GF/C membrane (W_1) and place in filter holder.
- iii) 100 ml sample after vigorous shaking is taken in measuring cylinder and is filtered through the membrane using a filtering apparatus. Continue suction until draining is complete.
- iv) Dry the membrane in an oven at 103 °C – 105 °C for one hour.
- v) Cool in a desiccator and weigh (W_2)

B) CALCULATION:

$MLSS \text{ (mg/l)} = (W_2 - W_1) \times 1000 \text{ Sample volume}$

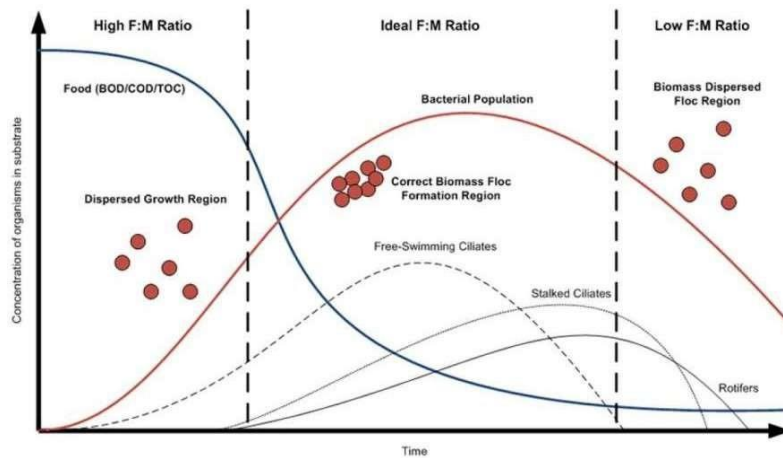
Where, W_2 = weight of membrane + MLSS

W_1 = weight of membrane

Ref: BIS 3025 Part 17

2. Food to Microorganism (F/M) ratio:

In order for the activated sludge process to operate properly there must be a balance between the food (BOD₅, COD, or TOC) entering the biological system, and the microorganisms in the aeration basin. A high F:M ratio means there is a greater quantity of food relative to the quantity of microorganisms available to consume that food. When the F:M ratio is high, the bacteria are active and dispersed and they multiply rapidly. But with a high F:M ratio the bacteria will not form a good floc. Operating the activated sludge process with a high F:M ratio will typically result in a poor settling sludge in the clarifier and a turbid effluent. A low F:M ratio means there are many microorganisms but there is a limited amount of food. Only when the food supply is limited do bacteria begin to develop a thicker slime layer, lose their motility, and begin to clump together to form floc that will settle well in the clarifier. Figure shows graphically how the bacterial population relates to the food supply. As you can see, a very high F:M ratio, or a very low F:M ratio will result in dispersed floc that will not settle well in the secondary clarifier.



There is no ideal F:M ratio that will work for all activated sludge treatment systems. Every wastewater treatment system is different, and each system has its own ideal or optimal F:M ratio. The best F:M ratio for a particular system depends on the type of activated sludge process and the characteristics of the wastewater entering the system. Generally, F/M ratio of 0.05 to 0.1 is used in extended aeration modification of activated sludge process.

The organic loading on the activated sludge aeration tank is evaluated in terms of Food to Microorganism ratio.

$$F/M = Q \times S_i / V \times X$$

Where, Q = Total Flow of Influent Sewage to Aeration Tank, (M³)/day

S_i = BOD₃ at 27 °C of Influent Sewage, mg/lit.

V = Volume of Aeration Tank (M³)

X = MLSS, mg/lit

3. Sludge Volume Index (SVI):

SVI is defined as the volume in ml occupied by one gram of activated sludge Mixed Liquor Suspended Solids (MLSS), dry weight, after settling of 30 min in a 1000 ml graduated cylinder. 1000 ml Mixed Liquor Suspended Solids (MLSS) from aeration tank is taken in graduated Imhoff cylinder and let the sludge settle for 30 minutes. Note the total volume occupied by settled sludge (V). The dry weight of the sludge is calculated as per MLSS method.

$$SVI = V \text{ in ml} / W \text{ in gm.}$$

Where V = Total Volume occupied by settled sludge in cylinder, ml

W = Dry weight of sludge in that volume, in gm.